8. Scaling an image to its bigger and smaller sizes.

Aim:

To perform basic image processing by scaling an image into bigger and smaller sizes using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2


img=cv2.imread(r"C:\Users\Maruthi\Downloads\METAL.jpg")


big=cv2.resize(img,None,fx=2,fy=2,interpolation=cv2.INTER_LI
NEAR)
small=cv2.resize(img,None,fx=0.5,fy=0.5,interpolation=cv2.INT
ER_AREA)


cv2.imshow("Original",img)
cv2.imshow("Bigger Image",big)
cv2.imshow("Smaller Image",small)
```

```
cv2.waitKey(0)

cv2.destroyAllWindows()
```

Sample Input

Sample Output

Result:

The input image was successfully scaled into bigger and smaller using the OpenCV library.